

Standard Operating Procedure: Pump Isolation & Maintenance

Document ID: SOP-COKE-OVEN-14. Version: 2.1.

Governing Equipment: Pump P-204, Pump P-205, Heat Exchanger HE-301.

This document describes the safety steps required before performing coupling alignment or bearing replacement on critical unit pumps.

Step 1: Electrical Isolation (LOTO) Isolate the primary breaker at switchgear panel CP-03. Apply physical lock and LOTO tag. Verify isolation by attempting a local start.

Step 2: Process Isolation Close suction valve HV-204A and discharge valve HV-204B. Open drain valve DV-204 to depressurize casing. Confirm pressure gauge reads 0 PSI before opening pump casing.

Step 3: Mechanical Alignment When replacing bearings, ensure coupling alignment is checked using the laser alignment tool. Misalignment of coupling must not exceed 0.05 mm.